AI-Driven Smart Agriculture & Crop Intelligence System

Using IoT (ESP32) and Machine Learning for Precision Farming

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Course: IoT

INTRODUCTION

- Agriculture often relies on manual observation for irrigation and crop care.
- This project introduces an IoT-based system that continuously monitors soil and environmental conditions.
- Machine Learning is used to convert sensor data into actionable crop intelligence.

WHY THIS PROJECT IS IMPORTANT?

- • Prevents over-irrigation and water wastage
- • Detects crop stress early
- • Enables data-driven farming decisions
- • Supports sustainable and precision agriculture
- • Suitable for farms and greenhouses

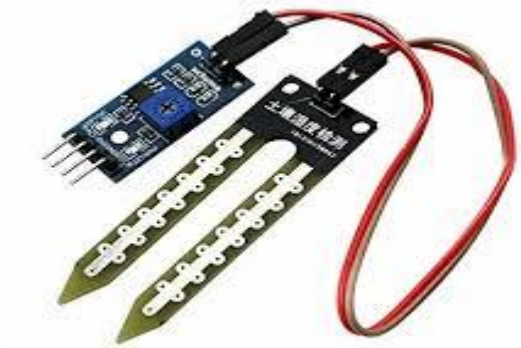
SYSTEM OVERVIEW

- Sensors collect real-time data → ESP32 processes and sends data to the cloud →
- Dashboard visualizes the data → Machine Learning models analyze the data to predict:
- • Irrigation needs
- • Crop health status
- • Yield trends

HARDWARE REQUIRED

- ESP32 DevKit V1 (ESP32-WROOM-32)

- Capacitive Soil Moisture Sensor



- DHT11/DHT22 Temperature & Humidity Sensor

- LDR Light Sensor

- Relay Module
- Breadboard and Jumper Wires
- Power Bank for field power supply

PIN CONNECTIONS (ESP32)

- Soil Moisture AO → GPIO 34
- LDR AO → GPIO 35
- DHT Data → GPIO 4
- Relay IN → GPIO 26
- VCC → 3.3V
- GND → GND

SOFTWARE & AI USED

- Arduino IDE for ESP32 programming
- Blynk IoT for cloud dashboard
- Python (Jupyter/Colab) for Machine Learning
- Decision Tree for crop health prediction
- Linear Regression for yield trend prediction

EXPECTED OUTCOME

- Live monitoring of soil and environment
- Smart irrigation alerts
- Crop health status prediction
- Yield trend analytics
- Portable battery-powered IoT field device